+

1'h0 CIN Add0

A[3..0] 4'h1 B[3..0] OUT[3..0] +

1'h0 CIN Add2

A[15..0] 16'h1 B[15..0] OUT[15..0] +

1'h0 CIN Add3

A[3..0] 4'h1 B[3..0] OUT[3..0] +

1'h0 CIN Add4

A[4..0] 5'h1d B[4..0] OUT[4..0] +

1'h0 CIN Add5

A[7..0] 8'h1 B[7..0] OUT[7..0] =Equal0

A[3..0] 4'hf B[3..0] OUT =Equal1

A[2..0] 3'h1 B[2..0] OUT =Equal2

A[2..0] 3'h2 B[2..0] OUT =Equal3

A[2..0] 3'h3 B[2..0] OUT =Equal4

A[2..0] 3'h4 B[2..0] OUT =Equal5

A[2..0] 3'h5 B[2..0] OUT =Equal6

A[2..0] 3'h6 B[2..0] OUT =Equal7

A[2..0] 3'h7 B[2..0] OUT =Equal8

A[7..0] 8'hc3 B[7..0] OUT

[1-2] F_SIMPAN <

1'h0 CIN LessThan1

16'h270f A[15..0] B[15..0] OUT

[1-2] OUT0

[1-2] OUT0

[1-2] OUT0

[1-2] OUT0

[1-2] OUT0

[1-2] OUT0

[1-2] OUT0

Selector0

SEL[2..0] DATA[2..0] OUT

Selector1

SEL[2..0] DATA[2..0] OUT

Selector2

SEL[2..0] DATA[2..0] OUT Selector3
SEL[2..0] DATA[2..0] OUT Selector4
SEL[2..0] DATA[2..0] OUT Selector5
SEL[2..0] DATA[2..0] OUT

Selector6

SEL[2..0] DATA[2..0] OUT

Selector7

SEL[2..0] DATA[2..0] OUT

Selector8

SEL[2..0] DATA[2..0] OUT

WideOr0 WideOr1 WideOr2

acc_terpilih[31..0] 01 acc_terpilih~[31..0] 01 acc_terpilih~[63..32] 01 acc_terpilih~[95..64] 01 acc_terpilih~[127..96] 01 acc_terpilih~[159..128] 01 acc_terpilih~[191..160] 01

ada_hasil D
CLK
SCLR

1'h0 Q

ada_hasil~0 0 1'h1 1 ada_hasil~1 0 1'h0 1

argmax10:u_amax

clk

in_valid mulai rst_n

in_acc[31..0]
in_idx[3..0] idx_maks[3..0]

awal D
CLK
SCLR

1'h0 Q

awal~0 0 1'h0 1 awal~1 0 1'h1 1

cacah_reset[3..0] D
CLK
ENA
SCLR

4'h0 Q

cacah_siklus[15..0] D
CLK
SCLR

16'h0 Q

cacah_siklus~[15..0] 01 cacah_siklus~[31..16] 0 16'h0 1 cacah_siklus~[47..32] 0 16'h0 1

citra_q[7..0] D
CLK
SCLR

8'h0 Q

citra

SYNC_RAM

WE 1'h0

ENA1 1'h1 CLR1 1'h0 DATAIN[7..0] 8'h0 WADDR[11..0] 12'h0 RADDR[11..0] DATAOUT[7..0] comb~[17..1] 01
comb~0

ctrl_fsm_p8:u_ctrl

clk epi_out_valid

rst_n
start

bias8[63..0]

s_ready_lajur[7..0] m_valid_lajur[7..0]

amax_valid
cfg_relu
cfg_round

done

epi_valid mem_en

s_first s_last s_valid
sibuk wr_en

addr_w[10..0] addr_d[8..0]

kel_bias[3..0]

s_init[255..0]

m_ready_lajur[7..0] pilih_lajur[2..0]

cfg_shift[4..0]

wr_addr[8..0]

amax_idx[3..0]

cuplik_d 0 1'h0 1 cuplik_d D

CLK
SCLR

1'h0 Q

epilog_unit:u_epi cfg_relu

cfg_round
clk
in_valid
rst_n

in_acc[31..0] cfg_shift[4..0] out_valid

out_act[7..0]

fase

awal clk

ctrl_fsm_p8:u_ctrl rst_n

start_r p[7..0]

tekan[2..0]

F_BANGUN
F_DIAM
F_JALAN
F_KE_DIAM
F_MUAT
F_SIMPAN
F_TUNTAS

idx[3..0] D
CLK
SCLR

4'h0 Q

[1-2]

idx_maks[3..0]

idx~[3..0] 01

idx~[7..4] 01

idx~[11..8] 0 4'h0 1

idx~[15..12] 01

idx~[19..16] 0 4'h0 1

k_lalu[2..0] D
CLK
SCLR

3'h0 Q

k_lalu~[2..0] 01

k_lalu~[5..3] 0 3'h0 1

mac_unit:g_lajur[0].u_mac 1'h0 cfg_relu 1'h0 cfg_round
clk m_ready
rst_n

s_first s_last s_valid
s_x[7..0] s_w[7..0]

s_init[31..0]

5'h0 cfg_shift[4..0] m_valid

s_ready m_acc[31..0]

mac_unit:g_lajur[1].u_mac 1'h0 cfg_relu 1'h0 cfg_round
clk m_ready
rst_n

s_first s_last s_valid
s_x[7..0] s_w[7..0]

s_init[31..0]

5'h0 cfg_shift[4..0] m_valid

s_ready m_acc[31..0]

mac_unit:g_lajur[2].u_mac 1'h0 cfg_relu 1'h0 cfg_round
clk m_ready
rst_n

s_first s_last s_valid
s_x[7..0] s_w[7..0]

s_init[31..0]

5'h0 cfg_shift[4..0] m_valid

s_ready m_acc[31..0]

mac_unit:g_lajur[3].u_mac 1'h0 cfg_relu 1'h0 cfg_round
clk m_ready
rst_n

s_first s_last s_valid
s_x[7..0] s_w[7..0]

s_init[31..0]

5'h0 cfg_shift[4..0] m_valid

s_ready m_acc[31..0]

mac_unit:g_lajur[4].u_mac 1'h0 cfg_relu 1'h0 cfg_round
clk m_ready
rst_n

s_first s_last s_valid
s_x[7..0] s_w[7..0]

s_init[31..0]

5'h0 cfg_shift[4..0] m_valid

s_ready m_acc[31..0]

mac_unit:g_lajur[5].u_mac 1'h0 cfg_relu 1'h0 cfg_round
clk m_ready
rst_n

s_first s_last s_valid
s_x[7..0] s_w[7..0]

s_init[31..0]

5'h0 cfg_shift[4..0] m_valid

s_ready m_acc[31..0]

mac_unit:g_lajur[6].u_mac 1'h0 cfg_relu 1'h0 cfg_round
clk m_ready
rst_n

s_first s_last s_valid
s_x[7..0] s_w[7..0]

s_init[31..0]

5'h0 cfg_shift[4..0] m_valid

s_ready m_acc[31..0]

mac_unit:g_lajur[7].u_mac 1'h0 cfg_relu 1'h0 cfg_round
clk m_ready
rst_n

s_first s_last s_valid
s_x[7..0] s_w[7..0]

s_init[31..0]

5'h0 cfg_shift[4..0] m_valid

s_ready m_acc[31..0]

mem_d:u_mem_d

clk

rd_en wr_en

rd_addr[8..0] wr_addr[8..0] wr_data[7..0]

rd_q[7..0]

mem_w:g_lajur[0].u_mem_w

clken

addr[10..0] q[7..0] mem_w:g_lajur[1].u_mem_w
clken

addr[10..0] q[7..0] mem_w:g_lajur[2].u_mem_w

clken

addr[10..0] q[7..0] mem_w:g_lajur[3].u_mem_w

clken

addr[10..0] q[7..0]

mem_w:g_lajur[4].u_mem_w

clken

addr[10..0] q[7..0] mem_w:g_lajur[5].u_mem_w

clken

addr[10..0] q[7..0] mem_w:g_lajur[6].u_mem_w

clken

addr[10..0] q[7..0]

mem_w:g_lajur[7].u_mem_w

clken

addr[10..0] q[7..0] p[7..0] D

CLK

SCLR

8'h0 Q

p_d[7..0] D
CLK
SCLR

8'h0 Q

p_d~[7..0] 0 8'h0 1 p~[7..0] 01 p~[15..8] 0 8'h0 1

rom_bias8:u_bias
kelompok[3..0] dout[63..0]

rst_n D
CLK
SCLR

1'h0 Q

siklus_r[15..0] D
CLK
SCLR

16'h0 Q

siklus_r~[15..0] 01 siklus_r~[31..16] 0 16'h0 1

siklus_tampil[13..0] 0 14'h270f 1

start_r 0 1'h0 1

start_r D
CLK
SCLR

1'h0 Q

tekan[0]

tekan[1]

tekan[2]

tulis_d D
CLK
SCLR

1'h0 Q

tulis_d~0 0 1'h0 1 0:7 3:0 3:0 6->DATA[0]25778 DATA[0]25831<-0 DATA[0]25786<-5 DATA[0]25794<-4 DATA[0]25807<-3 DATA[0]25823<-1 DATA[0]25815<-2 DATA[0]25770<-7 7:0

0:2 102

13:0 102 0:7

10:17
1:9 1->DATA[1]25779 DATA[1]25832<-7 DATA[1]25787<-2 DATA[1]25795<-3 DATA[1]25808<-4 DATA[1]25824<-6 DATA[1]25816<-5 DATA[1]25771<-0 01 2 3 4 5 6 7 0:7

31:0

63:32

95:64 127:96 159:128 191:160

223:192

255:224 0:8

+

1'h0 CIN Add1

A[31..0] 32'h1 B[31..0] OUT[31..0] +

1'h0 CIN Add6

A[3..0] 4'h5 B[3..0] OUT[3..0] +

1'h0 CIN Add7

A[3..0] 4'h3 B[3..0] OUT[3..0] +

1'h0 CIN Add8

A[3..0] 4'h3 B[3..0] OUT[3..0] +

1'h0 CIN Add9

A[3..0] 4'h3 B[3..0] OUT[3..0] +

1'h0 CIN Add10

A[3..0] 4'h3 B[3..0] OUT[3..0] +

1'h0 CIN Add11

A[3..0] 4'h3 B[3..0] OUT[3..0] +

1'h0 CIN Add12

A[3..0] 4'h3 B[3..0] OUT[3..0] +

1'h0 CIN Add13

A[3..0] 4'h3 B[3..0] OUT[3..0] +

1'h0 CIN Add14

A[3..0] 4'h3 B[3..0] OUT[3..0] +

1'h0 CIN Add15

A[3..0] 4'h3 B[3..0] OUT[3..0] +

1'h0 CIN Add16

A[3..0] 4'h3 B[3..0] OUT[3..0] +

1'h0 CIN Add17

A[3..0] 4'h3 B[3..0] OUT[3..0] +

1'h0 CIN Add18

A[3..0] 4'h3 B[3..0] OUT[3..0] +

1'h0 CIN Add19

A[3..0] 4'h3 B[3..0] OUT[3..0] +

1'h0 CIN Add20

A[3..0] 4'h3 B[3..0] OUT[3..0] +

1'h0 CIN Add21

A[3..0] 4'h3 B[3..0] OUT[3..0] +

1'h0 CIN Add22

A[3..0] 4'h3 B[3..0] OUT[3..0] +

1'h0 CIN Add23

A[3..0] 4'h3 B[3..0] OUT[3..0] +

1'h0 CIN Add24

A[3..0] 4'h3 B[3..0] OUT[3..0] +

1'h0 CIN Add25

A[3..0] 4'h3 B[3..0] OUT[3..0] +

1'h0 CIN Add26

A[3..0] 4'h3 B[3..0] OUT[3..0] +

1'h0 CIN Add27

A[3..0] 4'h3 B[3..0] OUT[3..0] +

1'h0 CIN Add28

A[2..0] 3'h3 B[2..0] OUT[2..0] +

1'h0 CIN Add29

A[3..0] 4'h3 B[3..0] OUT[3..0] +

1'h0 CIN Add30

A[3..0] 4'h3 B[3..0] OUT[3..0] +

1'h0 CIN Add31

A[3..0] 4'h3 B[3..0] OUT[3..0] +

1'h0 CIN Add32

A[3..0] 4'h3 B[3..0] OUT[3..0] +

1'h0 CIN Add33

A[31..0] 32'h1 B[31..0] OUT[31..0] Decoder0

IN[3..0] OUT[15..0] =Equal9

A[3..0] B[3..0] OUT

[1-2] F_SIMPAN <

1'h1 CIN LessThan0

32'h7a11f A[31..0] B[31..0] OUT <

1'h1 CIN LessThan2

4'ha A[3..0] B[3..0] OUT <

1'h1 CIN LessThan3

4'ha A[3..0] B[3..0] OUT <

1'h1 CIN LessThan4

3'h5 A[2..0] B[2..0] OUT <

1'h1 CIN LessThan5

4'h5 A[3..0] B[3..0] OUT <

1'h1 CIN LessThan6

4'h5 A[3..0] B[3..0] OUT <

1'h1 CIN LessThan7

4'h5 A[3..0] B[3..0] OUT <

1'h1 CIN LessThan8

3'h5 A[2..0] B[2..0] OUT <

1'h1 CIN LessThan9

4'h5 A[3..0] B[3..0] OUT <

1'h1 CIN LessThan10

4'h5 A[3..0] B[3..0] OUT <

1'h1 CIN LessThan11

4'h5 A[3..0] B[3..0] OUT <

1'h1 CIN LessThan12

4'h5 A[3..0] B[3..0] OUT <

1'h1 CIN LessThan13

4'h5 A[3..0] B[3..0] OUT <

1'h1 CIN LessThan14

4'h5 A[3..0] B[3..0] OUT <

1'h1 CIN LessThan15

3'h5 A[2..0] B[2..0] OUT <

1'h1 CIN LessThan16

4'h5 A[3..0] B[3..0] OUT <

1'h1 CIN LessThan17

4'h5 A[3..0] B[3..0] OUT <

1'h1 CIN LessThan18

4'h5 A[3..0] B[3..0] OUT <

1'h1 CIN LessThan19

4'h5 A[3..0] B[3..0] OUT <

1'h1 CIN LessThan20

4'h5 A[3..0] B[3..0] OUT <

1'h1 CIN LessThan21

4'h5 A[3..0] B[3..0] OUT <

1'h1 CIN LessThan22

4'h5 A[3..0] B[3..0] OUT <

1'h1 CIN LessThan23

4'h5 A[3..0] B[3..0] OUT <

1'h1 CIN LessThan24

4'h5 A[3..0] B[3..0] OUT <

1'h1 CIN LessThan25

3'h5 A[2..0] B[2..0] OUT <

1'h1 CIN LessThan26

4'h5 A[3..0] B[3..0] OUT <

1'h1 CIN LessThan27

4'h5 A[3..0] B[3..0] OUT <

1'h1 CIN LessThan28

4'h5 A[3..0] B[3..0] OUT <

1'h1 CIN LessThan29

4'h5 A[3..0] B[3..0] OUT <

1'h1 CIN LessThan30

32'h17d783f A[31..0] B[31..0] OUT
[1-2] OUT0

[1-2] OUT0

[1-2] OUT0

[1-2] OUT0

[1-2] OUT0

[1-2] OUT0

[1-2] OUT0

WideOr4 WideOr5 WideOr6
WideOr7

cacah_detak[31..0] D
CLK
SCLR

32'h0 Q

cacah_detak~[31..0] 0 32'h0 1

cacah_detak~[63..32] 0 32'h0 1

cacah_sampel[31..0] D
CLK
SCLR

32'h0 Q

cacah_sampel~[31..0] 0 32'h0 1

cacah_sampel~[63..32] 0 32'h0 1

clk

cuplik 0 1'h0 1 cuplik D

CLK
SCLR

1'h0 Q

detak D
CLK
SCLR

1'h0 Q

detak~0 01

detak~1 0 1'h0 1

dig[3..0]

hasil_r[3..0] D
CLK
SCLR

4'h0 Q

hasil_r~[3..0] 01 hasil_r~[7..4] 0 4'h0 1

[1-2]

idx_maks[3..0]

idx_satuan[3..1] 01

k_stabil[3..0] D
CLK
SCLR

4'h0 Q

k_stabil~[3..0] 01

k_stabil~[7..4] 0 4'h0 1

key[3..0] k1[3..0] D

CLK
SCLR

4'h0 Q

k2[3..0] D
CLK
SCLR

4'h0 Q

led[0]~not
led[1]~not

led[2]~not

led[3..0]

led[3]~not
led_tinggi[2]

led_tinggi[3]

seg[7..0]

seg7_scan:u_scan

clk
rst_n

nilai[15..0]

titik[3..0] seg[7..0] dig[3..0]

s~[3..0] 01

s~[7..4] 01

s~[11..8] 01

s~[15..12] 01

s~[19..16] 01

s~[23..20] 01

s~[27..24] 01

s~[31..28] 01

s~[35..32] 01

s~[39..36] 01

s~[43..40] 01

s~[47..44] 01

s~[51..48] 01

s~[55..52] 01

s~[59..56] 01

s~[63..60] 01

s~[67..64] 01

s~[71..68] 01

s~[75..72] 01

s~[79..76] 01

s~[83..80] 01

s~[86..84] 01

s~[90..87] 01

s~[94..91] 01

s~[98..95] 01

s~[101..99] 01

tampil[3..0] 01

tampil[4] 01

tampil[15..5] 01

15

1297

118640

110
141

15

11641

13

103 3:1
3:1 0->DATAA27466 3:1 11{3} 4{3} 3 titik[2]194<-3 3 3:0 15:5 13:11 13:11 11:13 10->B[0]26614 A[0]26621<-10 DATAA26640<-10 9->B[0]26648 A[0]26655<-9 9->DATAA26674 B[0]26682<-8 A[0]26689<-8 8->DATAA26708 7->B[0]26748 A[0]26755<-7 DATAA26774<-7 6->A[0]26823 B[0]26816<-6 6->DATAA26842 5->B[0]26884A[0]26891<-5 5->DATAA26910 4->B[0]26984 A[0]26991<-4 4->DATAA27010 A[0]27093<-3B[0]27086<-3 DATAA27112<-3 B[0]27188<-2 A[0]27195<-2 DATAA27214<-2 B[0]27317<-1 A[0]27324<-1 1->DATAA27343 0->DATAB27491 1:3 1:3 3:1 0->B[2]26717 A[2]26723<-0 DATAA26734<-0 5:7 5:7 7:5 4->B[1]26716 A[1]26722<-4 4->DATAA26737 8->B[0]26715 9:11 A[0]26721<-8 9:11 8->DATAA26740 11:9 12->A[0]26789 13:15 B[0]26782<-12 13:15 12->DATAA26808 15:13 17:19 17:19 19:17 A[2]26959<-16 B[2]26953<-16 DATAA26970<-16 20->B[0]26850 21:23 A[0]26857<-20 21:23 23:21 DATAA26876<-20 25:27 25:27 27:2524->B[1]26952 A[1]26958<-24 24->DATAA26973 A[0]26925<-28 29:31 29:31 B[0]26918<-28 31:29 DATAA26944<-28 32->B[0]26951 33:35 A[0]26957<-32 33:35 32->DATAA26976 35:33 36->B[0]27018 37:39 A[0]27025<-36 37:39 39:37 DATAA27044<-36 40->A[0]27059 41:43 41:43 B[0]27052<-40 40->DATAA27078 43:41 45:47 45:47 47:45 A[2]27297<-44 B[2]27291<-44 44->DATAA27303 48->B[0]27120 49:51 A[0]27127<-48 49:51 48->DATAA27146 51:49 52->B[0]27154 53:55 A[0]27161<-52 53:55 55:53 DATAA27180<-52 57:59 57:5959:57 A[1]27296<-56 B[1]27290<-56 56->DATAA27306 60->A[0]27229 61:63 61:63 B[0]27222<-60 63:61 DATAA27248<-60 64->A[0]27263 65:67 65:67 B[0]27256<-64 64->DATAA27282 67:65 69:71 A[0]27295<-68 69:71 B[0]27289<-68 71:69 DATAA27309<-68 73:75 B[0]27351<-72 A[0]27358<-72 73:75 72->DATAA27377 75:73 77:79 B[0]27385<-76 A[0]27392<-76 77:79 76->DATAA27411 79:77 81:83 B[0]27419<-80 A[0]27426<-80 81:83 80->DATAA27442 83:81 84:86 84:86 86:85 88:90 87 91:94 95:98 99:101